Glove Size Finder

Product build plan for the team

Prepared for	The two team members (Member 1: research and operations, Member 2: engineering) and the study owner
Date	20 October 2026, draft version 1 for discussion
Repository	github.com/ValGrow-Labs/OPEN-CV (branch main)
Product status	Working prototype; unvalidated. No real participant has been measured yet.

Summary

The software prototype works end to end, but its accuracy has never been checked on real hands. Finishing the product takes six phases in a fixed order: choose the capture method with a comparison study, build that method properly, validate it on held-out people with the client's real size chart, complete the product around it, and launch. Two members can do this: **Member 1** leads research and operations, **Member 2** leads engineering, and the **study owner** makes the decisions and signs off.

The most important rule: the product may not claim accuracy, or be called validated, until the held-out validation study in phase 3 meets the targets agreed with the client and a person signs it off.

Where we are today

Area	Status
Coin-on-palm measurement in the phone browser	Built. Shows a model-based uncertainty (about ±7–8 mm for hand length). Not validated.
Sizing engine	Built, with a synthetic demo chart. No real manufacturer chart yet. Gives no size when the scan cannot single one out.
Operator validation screen	Built: repeated reference readings, per-scan accuracy and running totals, blinded mode, stored records, reports.
Remote pilot (invite links)	Built. Volunteer invites stay locked until the consent text is reviewed.
Flat-hand-on-paper method	Experimental, tested on synthetic images only. No customer screen.
Comparison study protocol and tooling	Ready. Waiting for decisions, approved consent and equipment.
Calibration	Uncalibrated. Cannot become “validated” without held-out evidence and human sign-off.
Deployment	Frontend on Vercel, backend on Render (set up by yonus07). Operator tools are off there.
Automated tests	328 backend, 91 frontend and 11 browser tests pass. All use synthetic data.
Real participant data	None.

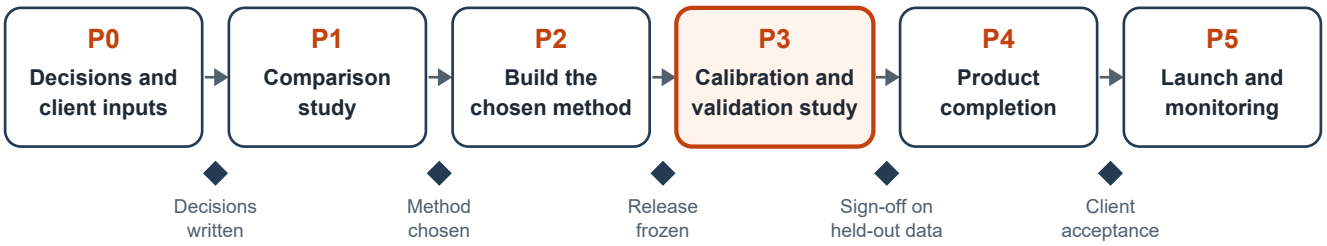
The first two weeks

These tasks do not depend on each other and can start today.

Member 1: research and operations	Member 2: engineering
Ask the client for the official size chart of every product, the measurement definitions behind it, the glove sizes available for try-on, and their accuracy targets.	Create the task board (one GitHub issue per task in this plan) and continuous integration that runs all tests on every pull request; protect the main branch.
Turn the consent draft (docs/CONSENT_REVIEW.md, section 4) into the final form; book the legal or ethics review.	Write the one-click start script for the study laptop (backend, HTTPS dev server and operator token).
Order a large sliding caliper or anthropometer (300 mm or more), a 150 mm caliper, a 100 mm check block, coins and A4 paper.	Test the operator screen on one Android phone and one iPhone over the laptop's HTTPS address; note any problem.
Prepare the room: matte dark table, even light, locked cabinet for forms.	If the study is in India: add local-coin selection with the ruler size check to the operator screen.
Practise the hand length and breadth measurements with the caliper on each other.	With yonus07: confirm the Vercel and Render builds pass and that no operator switch or token is set on Render.

Roadmap

Each phase ends at a gate (diamond). P4 engineering runs alongside P3.



Phase	Goal	Gate before the next phase	Lead	Rough duration*
P0 Decisions and client inputs	Settle location, targets, participant cap and consent owner; receive the real size chart and definitions.	All decisions written into the protocol; chart received.	Owner, Member 1	1–2 weeks
P1 Comparison study	Choose coin or paper with a blinded, randomised study on the same participants.	Analysis done; the owner's decision recorded.	Member 1, Member 2	4–8 weeks
P2 Build the chosen method	A production-quality version of the chosen method in the phone browser.	Frozen release candidate passes all tests on the target phones.	Member 2	2–6 weeks
P3 Calibration and validation	Fit on development participants, measure accuracy on held-out ones, with glove try-on.	Held-out results meet the targets; human sign-off recorded.	Member 1, Member 2	8–12 weeks
P4 Product completion	Store integration, accounts, languages, accessibility, security, monitoring.	Security and privacy review passed; client acceptance.	Member 2	4–8 weeks, alongside P3
P5 Launch and monitoring	Soft launch, measure fit outcomes and returns, improve.	Launch metrics agreed with the client are met.	All	Ongoing

* Rough estimates for planning only. Recruitment speed and approvals decide the real durations.

Roles and how we work

Role	Responsibilities
Study owner	Makes the decisions in P0, talks to the client, approves consent, signs off validation, makes the final calls. If possible, holds the sealed randomisation envelopes and does the second data entry.
Member 1: research and operations	Consent and legal, client inputs, recruitment, the caliper reference measurements (the measurer), try-on sessions, data entry, analysis together with Member 2 after collection.
Member 2: engineering	All software, testing, deployment and security; the scan operator during studies; data imports and the analysis scripts.

- **Blinding.** During data collection the measurer (Member 1) never sees camera output, and the scan operator (Member 2) never sees the reference measurements or any results. Results are looked at only after collection ends.
- **Main deploys automatically.** Pushing to main can redeploy the live site, so only tested work is merged, through a pull request reviewed by the other member.
- **Weekly 30-minute check-in** on the task board: done, blocked, next.

Detailed tasks by phase

P0 Decisions and client inputs

Task	Owner	Done when
Decide the study location: United States (works today) or India (needs the local-coin option first).	Owner	Decision written down.
Agree the smallest meaningful difference (proposed 1.5 mm), acceptable rejection and self-scan rates, and the participant cap (60, up to 120).	Owner with client	Values written into the comparison study protocol before any data is analysed.
Get the client's official size chart for each product and its measurement definitions (which wrist crease; circumference with or without the thumb).	Member 1	Files received and stored; definitions compared with ours.
Agree accuracy and size-accuracy targets for launch (for example hand length average error up to 3 mm, adjacent-size accuracy of 98% or more).	Owner with client	Targets signed by the client.
Name the consent approver, the study contact and the data deletion date.	Owner	Filled in on the consent form.
Book the legal or ethics review for each country (hand geometry can count as biometric data, for example under India's DPDP Act or US state laws such as Illinois BIPA).	Member 1	Reviewer and date confirmed.

P1 Comparison study: coin on palm vs hand flat on paper

Task	Owner	Done when
Finalise the comparison-study consent form and tick every item of the checklist in docs/CONSENT_REVIEW.md, section 5.	Member 1	Approved by the owner and the reviewer.
Buy and check the instruments every study day against the 100 mm block.	Member 1	Check logged on the case-report form.
Generate the randomisation list with a sealed seed and make one sealed envelope per participant.	Member 1 or owner	Envelopes ready; seed kept sealed.
Add the local-coin option to the operator screen (only if the study is in India).	Member 2	Tests pass; tried on a real phone.
Set up the study phone (cloud backup and location off) and laptop (BitLocker on, C:\study not synced).	Member 2	Set-up items of the data-protection checklist ticked.
Dry run with the two members as participants, recorded as Software test.	Both	Coin scan and paper photo work end to end in blinded mode; photos deleted.
Recruit and schedule 10–15 feasibility participants, then 60 fresh confirmatory participants.	Member 1	Sessions booked.
Run the sessions: Member 1 measures with calipers; Member 2 runs both methods in the envelope's order, retaking a scan only when the app rejects it.	Both	Every session's data and photo-deletion log complete.
After feasibility: enter the forms twice, re-estimate the sample size using spreads only, record changes, then freeze code and settings.	Both	Freeze record (commit, settings fingerprint) written into the protocol.
After the confirmatory phase: run the pre-specified analysis.	Both	Report delivered; the owner records the decision.

P2 Build the chosen method

Task	Owner	Done when
If coin is chosen: offer each country's verified coins in the customer flow with the ruler size check; improve on-screen guidance; speed up model loading on low-end phones.	Member 2	Release candidate passes all tests.
If paper is chosen: port the paper pipeline to the browser (sheet detection, perspective correction, hand outline, wrist alignment cue) so photos never leave the phone; add Python-vs-browser parity tests as for the coin detector.	Member 2	Browser results match Python on shared test images.
Build the customer screens for the chosen method; keep the tape-measure path for scans that cannot pick one size.	Member 2	Reviewed by Member 1 with three test users.
Device test on at least 6 phones (3 Android price tiers, 2 iPhones, 1 tablet) and the main browsers.	Member 2	Test matrix filled in; blocking issues fixed.
Freeze the release: tag the commit and record the settings fingerprint.	Member 2	Tag created; the same version is used for all of P3.

P3 Calibration and validation study

Task	Owner	Done when
Recruit at least 120 adults covering every glove size, sexes, skin tones and ages; split them into development and held-out test groups before any analysis.	Member 1	Split recorded; nobody in both groups.
Measure references with calipers (and a tape for circumference if the chart uses it); repeat each reading.	Member 1	Reference forms complete.
Scan each participant on at least two devices with the frozen release.	Member 2	Scans stored with their settings fingerprint.
Try-on: each participant tries the neighbouring sizes of the client's gloves; record the best and acceptable sizes.	Member 1	Fit trial file complete.
Fit the calibration on development participants only (validation/fit_calibration.py); evaluate on the held-out group (validation/evaluate.py).	Member 2	Held-out report with error, bias, coverage and size accuracy.
Compare with the targets. Only if they are met does a person add the sign-off block to config/measurement_calibration.json.	Owner	Sign-off recorded, or the product stays unvalidated.

P4 Product completion (alongside P3)

Task	Owner	Done when
Load the client's real size charts, one per product, and remove the demo-chart label from the customer flow.	Member 2	Charts validated at startup; client checked the sizes.
Integrate with the client's store: an embeddable widget or link that opens with the product, and passes the size back to the product page.	Member 2	Works on the client's test store.
Replace the shared operator token with real accounts, roles and an audit log for operator tools.	Member 2	Each operator logs in with their own account.
Security: rate limits, strict CORS, dependency audit, secrets only in Vercel and Render settings, external review.	Member 2	Review findings fixed.
Privacy: privacy notice, consent screens for any data collection, data retention jobs, a data-protection impact assessment.	Member 1	Approved by the legal reviewer.
Monitoring: error tracking, uptime checks and anonymous counts of scans started, finished and rejected (no personal data).	Member 2	Dashboard live.
Accessibility (WCAG 2.2 AA) and the languages the client needs.	Member 2	Audit passed; translations reviewed by native speakers.
Write the customer FAQ, support guide and operator manual.	Member 1	Published with the release.

P5 Launch and monitoring

Task	Owner	Done when
Soft launch to a small share of the client's customers.	Owner	Launch date agreed with the client.
Track scan completion, rejection rate, size distribution and glove returns or exchanges for fit, against the period before launch.	Member 1	Monthly report to the client.
Re-validate after any change to the algorithm or thresholds (a new settings fingerprint means a new validation).	Member 2	No unvalidated change reaches customers.
Only claim accuracy within the validated scope (devices, countries, products tested).	Owner	Marketing text checked against the validation report.

What we need from the client

Input	Why we need it	Needed by
Official size chart per product	The demo chart is invented; sizes must come from the manufacturer.	P0
Measurement definitions behind the chart	Our measurement must match theirs (wrist crease, circumference).	P0
Accuracy and size-accuracy targets	Decide when the product is good enough to launch.	P0
Gloves in neighbouring sizes for try-on	Size accuracy can only be measured with real fittings.	P3
Countries, coins and languages	Decide which coins and translations to support.	P2
Store platform and contact	Plan the website integration.	P4
Branding and hosting domain	Replace the neutral placeholder design.	P4

Risks

Risk	Effect	What we do about it
Recruitment is slow	Every phase moves later	Start recruiting in P0; book sessions in blocks; both members recruit.
Neither method reaches the targets	No validated product	Keep the tape-measure path; try improvements on development data only; report honestly.
Consent or legal review takes long	P1 cannot start	Book the review in the first two weeks.
A push to main breaks the live site	Customers see errors	Pull requests, tests in CI, protected main.
Wrong coin series (India)	About 17% scale error	Ruler size check; one verified coin per country.
Data-protection incident	Legal and trust damage	Checklists, encryption, photos deleted after processing, no names in data files.
Two people for many tasks	Slippage	P4 work only on non-session days; the owner helps with envelopes and data entry.

Rules for everyone

- Never create, edit or guess measurement data. Synthetic test results are never reported as accuracy.
- Keep blinding during collection; look at results only after collection ends.
- Do not change config/measurement_calibration.json or mark anything validated without the owner's sign-off on held-out data.
- No customer images leave the phone. Study photos are deleted right after processing, and the deletion is logged.
- No names, phone numbers or emails in any data file; only random participant codes.
- Merge to main only through a reviewed pull request with all tests passing.

Reference

Document or command	What it is
docs/COMPARISON_STUDY_PROTOCOL.md	Full protocol for P1: definitions, blinding, sample size, analysis plan.
docs/CONSENT_REVIEW.md	Consent audit, draft comparison-study consent, data-protection checklists.
docs/VALIDATION_PLAN.md	Plan for P3: split, metrics, acceptance targets.
docs/PILOT_PROTOCOL.md	In-person and remote pilot procedures, measurement steps.
docs/MEASUREMENT_METHODOLOGY.md	How the coin method measures and its known limitations.
docs/PAPER_METHOD_EVALUATION.md	The paper method: design, assumptions, synthetic results.
python -m validation.comparison randomise | import-coin | process-paper | nuisance | analyse	Study tooling for P1 (run from backend/).
python -m scripts.validation_report --records	Accuracy report from validation-mode records.
pytest, npm test, npm run e2e	Automated tests (backend, frontend, browser).